# Omkar Chandra Ramachandra

Postdoctoral Scholar, Opal Lab, Department of Neurology
Northwestern University Feinberg School of Medicine
Publishes as Omkar Chandra
omkarchandra.ramachandra@northwestern.edu
ORCID 0009-0002-3143-5464 · Google Scholar

## Research Interests

Modelling living systems as coupled dynamical layers (gene expression, proteome, bioelectric state) to find the rules that hold a phenotype in place and the interventions that move it.

## Research Experience

Feb 2025 – present — **Postdoctoral Scholar, Opal Lab, Department of Neurology, Northwestern University Feinberg School of Medicine.**
Genomic data analysis, time-series data modelling, training in patch-clamp and cell-attached recording.
Reanalysis of longitudinal SCA1 single-nucleus RNA-seq data to characterise the Purkinje-cell disease state, and transcriptome-informed Purkinje-cell conductance modelling (ongoing).

2018 – 2024 — **Ph.D. Researcher, IIIT Delhi** (Advisor: Vibhor Kumar). Interpretable machine-learning models of regulatory logic at promoters and single-cell chromatin methods.

2017 – 2018 — **Research Associate, Epigenomics Lab, AIIMS Delhi.**
*Role of non-coding RNAs in ovarian cancer.*

## Education

2018 – 2024 — **Joint Ph.D.–M.Tech. programme in Computational Biology, IIIT Delhi.**
Ph.D. thesis: *Explainable machine learning with epigenomic features for insights into regulatory and functional genomics* (advisor Vibhor Kumar; defended 16 December 2024; graduation requirements completed in Monsoon 2024).
M.Tech. coursework 2018–2019; thesis December 2022 (supervisors Vibhor Kumar and Say Li Kong): *Designing Circulating-tumor cells (CTCs) detection method using co-measurement data of Single-Cell RNA and Targeted DNA Sequencing*.

2014 – 2016 — **M.Sc. Biochemistry, University of Mysore**
Thesis: *Purification of an anti-haemorrhagic protein from Indian cobra Naja naja venom* (2016 project work).
Dissertation (2015): *Sterol Regulatory Element Binding Proteins*.

2011 – 2014 — **B.Sc. (Biochemistry, Microbiology and Biotechnology), Yuvaraja's College, University of Mysore**

## Research Publications

### Peer-reviewed articles

1. **Chandra O**, Sharma M, Pandey N, Jha IP, Mishra S, Kong SL, Kumar V, "Patterns of transcription factor binding and epigenome at promoters allow interpretable predictability of multiple functions of non-coding and coding genes," *Computational and Structural Biotechnology Journal*, vol. 21, pp. 3590–3603, 2023. doi: 10.1016/j.csbj.2023.07.014.
2. Mishra S, Pandey N, Chawla S, Sharma M, **Chandra O**, Jha IP, SenGupta D, Natarajan KN, Kumar V, "Matching queried single-cell open-chromatin profiles to large pools of single-cell transcriptomes and epigenomes for reference supported analysis," *Genome Research*, vol. 33, no. 2, pp. 218–231, 2023. doi: 10.1101/gr.277015.122.
3. Sharma M, Jha IP, Chawla S, Pandey N, **Chandra O**, Mishra S, Kumar V, "Associating pathways with diseases using single-cell expression profiles and making inferences about potential drugs," *Briefings in Bioinformatics*, vol. 23, no. 4, 2022, Art. no. bbac241. doi: 10.1093/bib/bbac241.
4. Pandey N, **Chandra O**, Mishra S, Kumar V, "Improving chromatin-interaction prediction using single-cell open-chromatin profiles and making insight into the cis-regulatory landscape of the human brain," *Frontiers in Genetics*, vol. 12, 2021, Art. no. 738194. doi: 10.3389/fgene.2021.738194.

### Preprints

1. **Chandra O**, Pramanik D, Gautam S, Sharma M, Dubey N, Mahato B, Orlov YL, Kumar V, "Explainable models using transcription factor binding and epigenome patterns at promoters reveal disease-associated genes and their regulators in the context of cell-types," *bioRxiv*, 2024, preprint, v2. doi: 10.1101/2024.05.06.592622.

2. Pandey N, Sharma M, Mathur A, Anene-Nzel CG, Hakimullah M, Patel P, Jha IP, **Chandra O**, Mishra S, Bhattacharya J, Sharma A, Foo R, Sandhu K, Mandoli A, DasGupta R, Kumar V, "Deciphering drug response and phenotypic heterogeneity of cancer cells using gene ensembles of regulatory units defined by chromatin domains," *bioRxiv*, 2023, preprint, v3. doi: 10.1101/2023.01.15.524115.

## Manuscripts in Preparation and Under Review

Datta S, Edamakanti CR, Kumar D, Smith CM, Mohan V, **Chandra O**, Panja C, Ghasempour Ramezan G, Surana P, Novakovic M, Kim H, Do J, Prakriya M, Martina M, Davuluri RV, Opal P. Astrocyte dysfunction disrupts synaptic regulation and cerebellar circuit activity in spinocerebellar ataxia type 1. Under revision, *Journal of Clinical Investigation*.

**Chandra O**, Phua VL-Y, Ling WH, Zhai W, Tan DMY, Tai DW-M, Lim TKH, Tam WL, Agarwal A, Kumar V, Kong SL. Concurrent single-cell transcriptome and targeted DNA sequencing enables marker-free detection of pancreatic circulating tumor cells and reveals a recurrent ciliated-epithelial program. In preparation; manuscript available on request.

**Chandra O**, Caleb M. Smith et al. (author list to be finalized). A transcript signature is not a conductance signature: the SCA1 Purkinje channel changes fail to reproduce irregular firing in two published biophysical models. In preparation.

## Service

Ad hoc reviewer: BMC Genomics (2025); npj Aging (2025)

## Software and Ongoing Projects

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| sca1-purkinje-state-modeling | Analysis and modelling code for the SCA1 Purkinje-cell project: single-nucleus state analysis and biophysical Purkinje-cell simulations (in development). |
| somaticseed | Tumour-cell identification from single-cell RNA-seq using RNA-derived somatic mutations and positive–unlabelled learning (in development). |
| ocdeck | Terminal operations console for AI coding agents. ocdeck on GitHub. |

## Skills

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| Genomics | snRNA-seq and scRNA-seq, scATAC-seq, trajectory and cell-state transition analysis, Oxford Nanopore long-read sequencing, repeat-expansion genotyping, methylation calling and phasing |
| Machine Learning | Interpretable models (random forest, XGBoost, SVM, lasso and ridge), positive–unlabelled learning, feature attribution |
| Engineering | Python, R, shell scripting, SLURM and high-performance computing, Git, LaTeX; agentic coding workflows |
| Wet lab | Molecular biology and biochemistry experiment design; bench electrophysiology (in training). |
| Languages | English, Kannada, Hindi. |
| Modelling & Theory | Boolean and gene-regulatory networks, biophysical neuronal modelling (NEURON), time-series analysis. |

## Awards

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| 2016 | CSIR-UGC NET Junior Research Fellowship, All-India Rank 133 (national doctoral fellowship, India); GATE qualified (Graduate Aptitude Test in Engineering). |

## References

**Dr. Puneet Opal, MD, PhD**
Professor of Neurology and Cell & Developmental Biology
Director, Denning Ataxia Center
Northwestern University Feinberg School of Medicine
(Current postdoctoral supervisor)
p-opal@northwestern.edu

**Dr. Vibhor Kumar**
Associate Professor
Department of Computational Biology
IIIT Delhi, New Delhi
Ph.D. advisor
vibhor@iiitd.ac.in

**Dr. Jaime García-Añoveros, PhD**
Professor of Anesthesiology, Neuroscience, and Neurology
Northwestern University Feinberg School of Medicine
Collaborated on a project in his lab
anoveros@northwestern.edu

**Dr. Say Li Kong**
Manager, Thermo Fisher Scientific
Singapore
M.Tech. thesis co-supervisor
sayli.kong@thermofisher.com